

ĐẠI HỌC QUỐC GIA THÀNH PHỐ HỒ CHÍ MINH
TRƯỜNG ĐẠI HỌC CÔNG NGHỆ THÔNG TIN
KHOA KHOA HỌC MÁY TÍNH



SELF-STUDY

Đề tài: WORD EMBEDDING

Mentor: Myself
Student: Lại Khánh Hoàng

Tp. Hồ Chí Minh



Mục lục

1	Why word embedding ?	2
2	Measuring word embeddings	3
3	Word Embedding Models	3



1 Why word embedding ?

In order to fit our text data into a neural network, we need to first encode it, meaning transfer these words to numbers. One simple way to implement this is to use one-hot vectors, each vector represents a specific word.

$$\text{I love eating} \rightarrow \text{Dict}(3) = \{I, \text{love}, \text{eating}\} \rightarrow \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \rightarrow \text{To model}$$

This works fine if we have a small vocabulary. But what if we are processing a text with thousands of words (which is usually the case) ? This is **not memory efficient and losing the context** (no relation between words).

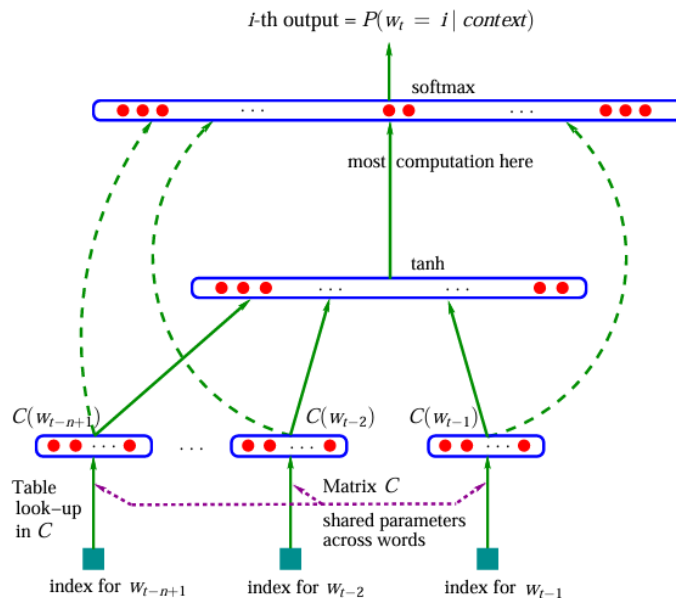
So we turn our words into a vector of values, each words as a dense vector of real numbers. Each value of the vector may represent a feature (chứa thông tin ẩn của từ, ví dụ: thích ngủ, thích uống trà sữa,...). Word embeddings encode semantic information.

Each input integer is used as the index to access a table that contains all possible vectors. All we need to specify the number of vocabulary and the size of embedding, example:

$$\begin{cases} \text{Nice to meet you} \\ \text{Great to meet you} \end{cases} \rightarrow \text{Dict}(5) = \{\text{nice}, \text{to}, \text{meet}, \text{you}, \text{great}\} \rightarrow \begin{cases} [0, 1, 2, 3] \\ [4, 1, 2, 3] \end{cases} \text{ join with } \begin{pmatrix} 0 & 0.6 & 0.8 \\ 1 & 0.7 & 1.3 \\ 2 & 0.5 & 1 \\ 3 & 0.9 & 0.4 \\ 4 & 0.4 & 0.9 \end{pmatrix}$$

Embedding: (5,2). 5 represent for the vocab size; and 2 represent for the embedding size, maybe stand for "Eating" and "Studying" of user.

After training the model, the network has learned the mappings. We can access any word vector by its index.



How is this done ? Using embedding layer (a layer that generates word embedding by multiplying an index vector with a word embedding matrix) - size: [Vocab size x Embedding size](#)



2 Measuring word embeddings

How are word vectors measured ?

$$\text{similarity}(A, B) = \cos(\theta) = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \|\mathbf{B}\|} = \frac{\sum A_i B_i}{\sqrt{\sum A_i^2} \sqrt{\sum B_i^2}}$$

This metric can be used for clustering words: words with similar meanings (e.g., study, studying, studied) will have embeddings that are close together in vector space.

3 Word Embedding Models

- Continuous bag-of-words (CBOW): Uses words before and after the target word to predict it (uses surroundings of the target word). Rather than feeding n previous words into the model, the model receives n words around the target word.
- Skip-Gram model: Opposite of CBOW, uses the center word to predict the surrounding words.